

Active sensing is not communication

Causal auditing of emergent signals in embodied multi-agent reinforcement learning, and the channel factorisation that lets signalling appear

Abstract

Every electric organ discharge (EOD) a weakly electric fish produces does two things at once: it illuminates the emitter’s own surroundings, and it broadcasts the emitter’s presence to anyone with the receptors to detect it. This dual use makes it unusually hard to say whether an EOD is *communication*. A recent multi-agent reinforcement learning model of electric fish collectives showed that EODs shape group foraging and spacing, and defined communication operationally as an EOD that conspecifics detect and respond to. That definition cannot separate a signal from a cue, and the single intervention available in that model—silencing one fish—removes the emitter’s own electrolocation and its neighbours’ access to it simultaneously, so the resulting behavioural change cannot be attributed to either. We build a GPU-vectorised reimplementation of that simulator, validated against the original to floating-point round-off (maximum relative deviation 6×10^{-10} across all sensory pathways), that runs 582k agent-steps per second and lets us switch the consequences of a discharge independently. A single motor act is decomposed into three separately controllable channels: *reaffERENCE* (the emitter’s own image of the world), *illumination* (the same pulse lighting up a neighbour’s world—an exploitable cue), and *detection* (the pulse registering on a neighbour’s knollenorgans—a candidate signal). We further add what the original model lacks: a metabolic cost per discharge, electroreceptive predators that home in on EODs, switchable cooperative and competitive food economics, tunable social range, and persistent individual identity. Silencing a fish costs it -0.52 food items per episode, of which -0.51 is the loss of its own electrolocation and only 0.03 the loss of its detectability by others. The public channel nonetheless satisfies both standard criteria for communication: it carries decodable information about food ($\Delta R^2 = 0.156$) and it moves receiver policies $13\times$ more than sensory noise does. Yet replaying, scrambling or fabricating it leaves both parties’ payoffs unchanged within our resolution; metabolic cost, eavesdropping predation and cooperative harvesting regulate discharge rate without making the pulse train more informative (per-pulse informativeness falls from 0.127 to 0.087); and once reception is held fixed so that only audibility is cut, emission policy is unchanged (sender shaping 0.040). Two standard diagnostics fail outright: positive signalling is *higher* in populations whose receivers are deaf (0.152 versus 0.086), and causal influence *rises* 12-fold as the channel becomes less informative. A system can pass every standard diagnostic for emergent communication and not be communicating. In a positive control that frees one bit of discharge subtype from the sensory function while leaving the pulse itself untouched, a payoff-relevant social channel does appear—in one of four ecological settings, the one in which private information is scarcest. Together these results show that standard emergent-communication diagnostics can misclassify an exploited sensory cue as communication, and identify channel factorisation—rather than ecological pressure—as what lets a distinct social function appear.

1 Introduction

A weakly electric fish navigates by emitting brief electric pulses and reading the distortions that nearby objects impose on the resulting field [1, 2]. The same pulses propagate far beyond the range at which they are useful for electrolocation, and other fish—equipped with dedicated knollenorgan electroreceptors tuned to conspecific discharges—can detect them at up to a metre [3]. An EOD is therefore simultaneously a private sensory probe and a public event. Which of these functions explains why the fish emits when it does is the central question of electric-fish neuroethology, and it is an instance of a much older problem in animal communication: distinguishing a *signal*, whose production has been shaped by its effect on receivers, from a *cue*, which receivers merely exploit [4, 5].

Singh et al. [6] introduced a computational framework in which agents with biophysically grounded electrosensory and electromotor systems are trained by multi-agent reinforcement learning to forage collectively. Their agents reproduce several features of real fish, and interventions on the model—ablating electroreceptor classes, silencing an agent’s EOD, and altering the food distribution—show that discharges influence group foraging and social spacing. They define communication operationally, as “EOD signals that are sensed by conspecifics via Knollenorgans and that influence their behavior.”

That definition is a statement about effects, and effects are cheap. The emergent-communication literature has repeatedly shown that behavioural association between one agent’s output and another’s behaviour is not sufficient evidence of signalling: a receiver may be responding to something correlated with the message, and a sender’s output may be informative without having been shaped by that informativeness at all [7, 8]. Lowe et al. [7] formalise the two requirements as *positive signalling* (the message carries information about the sender’s state) and *positive listening* (the message causally changes the receiver’s behaviour), and note that either can occur without the other.

For electric fish the problem is sharper than in a generic emergent-communication setup, because the channel is not a dedicated cheap-talk symbol but a physical act with an unavoidable private benefit. Three consequences follow from a single discharge:

1. **Reafference.** The pulse casts an electric image of nearby objects onto the emitter’s own mormyromasts. This is active electrolocation and is the reason the behaviour exists at all.
2. **Illumination.** The same field polarises objects in a neighbour’s vicinity, and the neighbour’s mormyromasts can read those images—collective electrosensing, demonstrated in real fish by Pedraja and Sawtell [9]. The neighbour gains information from a probe it did not pay for. This is the paradigm case of a *cue*.
3. **Detection.** The pulse registers on the neighbour’s knollenorgans, which are dedicated to conspecific discharges and encode the emitter’s identity and bearing. This is the only one of the three that could be a *signal* in the strict sense.

Silencing a fish removes all three at once. Any behavioural consequence is therefore uninterpretable: it may be the muted fish foraging worse because it can no longer see, its neighbours foraging worse because they lost a source of free illumination, or its neighbours behaving differently because a signal they were attending to has disappeared. This confound is not specific to the model; it is the reason EOD-ablation experiments in real fish are hard to interpret.

Contributions. We make the three consequences independently switchable and ask what changes.

1. A **channel algebra** for dual-use signals. Reafference, illumination and detection become three binary masks applied to one motor act, letting us construct pulses that are private, exploitable-but-undetectable, detectable-but-uninformative, phantom (heard though never emitted), replayed from a different context, or scrambled in time or sender identity.
2. An **interventional** measure of causal influence. Because we own the simulator, the counterfactual in Jaques et al. [8] can be evaluated by the do operator directly—re-rendering the receiver’s sensory input under $\text{do}(\text{emit}_j=1)$ and $\text{do}(\text{emit}_j=0)$ with every other cause held on its factual path—rather than approximated with a learned model of other agents.
3. A **cue/signal discriminator**. Following Scott-Phillips [5], a signal requires that production be shaped by its effect on receivers. We train matched populations in worlds whose only difference is whether receivers can hear at all, and measure the divergence between the resulting emission policies against the seed-to-seed divergence within each world—a Sender Shaping Index (SSI).
4. An **economics** for the discharge. The original reward has no EOD term, so pulses are free. We add a metabolic cost per discharge, predators that localise prey by their discharges, cooperative versus competitive harvests, tunable social range, and persistent identity, and map the resulting regimes.
5. A **positive control**. Because a negative result is only as strong as the positive control beside it, we also give the agents a signalling variable that is *partially decoupled* from sensing—a discharge subtype that rides on the pulse, so sending still costs a pulse, but whose value changes nothing about the emitter’s own electric image, nothing about how the pulse illuminates neighbours, and nothing about predator

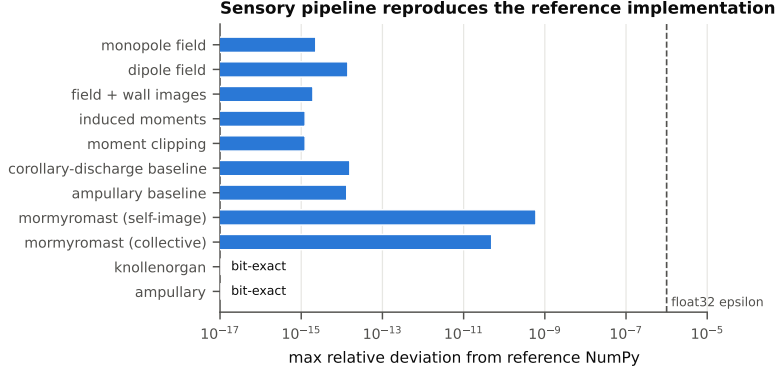


Figure 1: The GPU reimplemention reproduces the reference simulator. Maximum relative deviation from the reference NumPy implementation over random scenes, in double precision. Bars at the left edge are exactly zero.

detectability. This is the manipulation that distinguishes “pressure does not create signals here” from “signals cannot be measured here”.

A note on wording. Our populations adapt within a training run by policy-gradient learning under a fixed reward; we do not run an evolutionary outer loop over heritable traits. We therefore write *emerge under learning* throughout, and reserve claims about evolutionary history for the Discussion, where we say explicitly what our design can and cannot license.

2 Results

2.1 A validated, fast reimplemention

Before any claim about communication, the physics has to be right. We checked every stage of the sensory pipeline against the reference NumPy implementation in double precision on random scenes (Fig. 1, Table 1). All 11 checks agree to floating-point round-off, with a maximum relative deviation of 6×10^{-10} ; the knollenorgan and ampullary pathways agree bit-for-bit. Batching thousands of arenas on GPU gives 582k agent-steps per second in aggregate on two GPUs, which is what makes 270 independent training runs affordable.

One incidental finding is worth recording for users of the reference code. Its configuration file specifies a wall-reflection coefficient of 0.95, but the scene builder never forwards it, so the published results in fact use unattenuated first-order images. Matching the configuration file rather than the code changes the ampullary readings qualitatively, because those are computed as a small difference between large numbers. We match the code.

2.2 The discharge is worth a great deal — to the emitter

Agents that can discharge eat 5.01 food items per episode; agents permanently muted from the start of training eat 1.78 (Fig. 2A; $2.8\times$, Hedges’ $g = 4.3$, $p = 0.0001$). The probe is doing real work, which is the precondition for everything that follows: an EOD is not a cheap-talk token, it is an expensive and useful act of sensing that happens to be public.

Removing the three consequences one at a time separates that value. Training without knollenorgan detection costs little (4.71 items); training without the ability to illuminate neighbours costs somewhat more (4.31); a purely private probe that neither illuminates nor is detected gives 3.92; and removing the emitter’s own refference while leaving the public consequences intact gives 4.27. The ordering already suggests where the value lies, but from-scratch comparisons confound the value of a channel with the difficulty of learning without it, so we turn to interventions on frozen policies.

Table 1: Agreement between the GPU reimplementation and the reference NumPy implementation, in double precision on random scenes. Error is the maximum absolute deviation normalised by the largest magnitude present.

Stage	Max. relative deviation
Coulomb field of point charges	2.3×10^{-15}
Field of point dipoles	1.4×10^{-14}
Field including first-order wall images	1.9×10^{-15}
Dipoles induced on conductors	1.2×10^{-15}
Charge-limiting of fish body moments	1.2×10^{-15}
Corollary-discharge baseline	1.6×10^{-14}
Ampullary intrinsic baseline	1.3×10^{-14}
Mormyromast, self-image mode	6.0×10^{-10}
Mormyromast, collective-sensing mode	4.9×10^{-11}
Knollenorgan (conspecific detection)	exact (0)
Ampullary (passive DC)	exact (0)

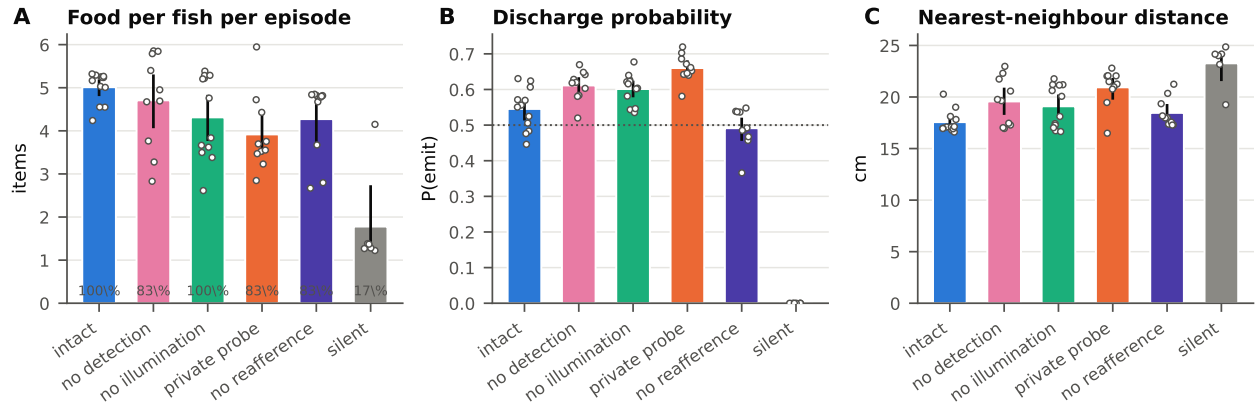


Figure 2: Removing each consequence of a discharge, trained from scratch. Bars are means over independent seeds with bootstrap 95% CIs; points are seeds. Percentages in (A) give the fraction of seeds that reached the learning criterion.

We note a methodological point that bears on how these numbers should be read. Training is bimodal: a seed either discovers patch foraging or remains near the floor set by passive sensing alone. The percentage under each bar is the fraction of seeds that cleared a fixed criterion (3 items per episode, applied identically to every condition). Conditions that remove sensory channels are not only worse on average, they are less reliably learnable, and we report both.

2.3 The muting effect is almost entirely private sensing

The central experiment. We take a trained, intact population and, at evaluation time only, silence one fish—the intervention used upstream—then repeat with each of its three consequences removed separately. Every condition is run from the same episode seed, so arena k has the same initial food layout and the same sensory-noise stream in all of them, and differences are taken within arena.

Muting one fish costs that fish -0.52 food items per episode (Fig. 3A). Decomposed: removing only its own refference costs -0.51, removing only its illumination of neighbours costs -0.00, and removing only its detectability costs 0.03. The private share accounts for essentially the whole effect.

The effect on its *neighbours* is smaller still (Fig. 3B): -0.01 items in total, with the illumination share 0.03 ($p = 0.5220$) and the detection share 0.01 ($p = 0.7234$). Group spacing (Fig. 3C) is likewise unchanged within resolution.

This is the paper’s first substantive result, and it is a negative one. When a fish is silenced in this model, what happens is overwhelmingly that the fish has been blinded, not that a conversation has been interrupted. An experiment that reports only the joint effect—as the reference study does, and as EOD-

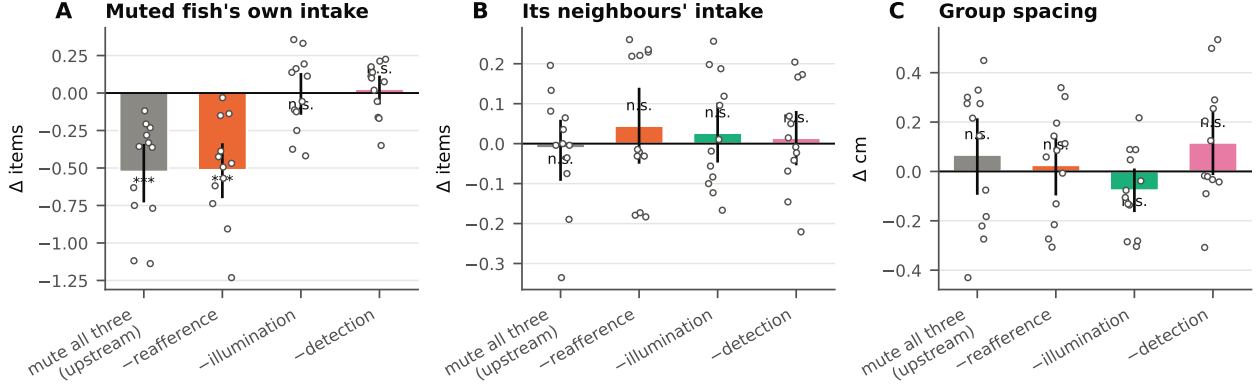


Figure 3: Silencing one fish removes three things at once; separating them reassigns the effect. Within-arena paired differences from the intact channel, frozen policies. Significance is a permutation test over seeds.

ablation experiments in real fish must—will attribute to communication an effect that is almost entirely private electrolocation.

2.4 The channel nonetheless carries information and moves policies

It would be wrong to conclude that the public channel is inert. It is not.

Discharge probability depends on the emitter’s situation: 0.586 when food is within 10 cm versus 0.457 when it is not ($p = 0.0001$; Fig. 4C). That state-dependence is inherited by the public channel, and a receiver can read it: decoding *only* from the number of pulses a receiver heard from a given emitter in a trailing window, food proximity is recovered with $\Delta R^2 = 0.156$ above a time-shifted baseline ($p = 0.0000$; Fig. 4B). Positive signalling about food, measured as mutual information conditioned on distance-to-neighbour decile so that geometry cannot manufacture it, is 0.086 of the state entropy (Fig. 4A).

Positive listening holds too. The interventional causal influence of Eq. 6—re-rendering a receiver’s sensory input under $\text{do}(e_j=1)$ versus $\text{do}(e_j=0)$ with everything else held on its factual path—is 0.0026 nats, against a noise floor of 0.00019 nats obtained by re-rendering the *same* intervention under an independent noise draw (Fig. 5A). The channel moves receiver policies $13\times$ more than sensory noise does.

Both of these diagnostics, however, behave badly here in ways worth recording. Positive signalling is *higher* in populations whose receivers are deaf (0.152) than in populations whose receivers can hear (0.086): a fish modulates its probe rate with its own situation whether or not anybody is listening, so the measure is picking up the sensory schedule, not signalling. This is exactly the artifact Lowe et al. [7] warn of, and it is straightforward to demonstrate once a deaf population is available as a control. Causal influence misbehaves in the other direction: it *rises* 12-fold, from 0.0026 to 0.0303 nats, as metabolic cost is increased—that is, precisely as the pulse train becomes rarer and less informative (Fig. 5D). A rare discharge is more surprising, so switching it on or off moves the receiver’s policy more, even though the train as a whole now tells the receiver less. Influence per message and information per unit time are different quantities, and in this system they are anti-correlated.

2.5 ... but it has no payoff consequence

Positive signalling and positive listening are both satisfied, so on the effect-based definition this is communication. The payoff test says otherwise.

Replacing an emitter’s public train with the train the same policy produced in a *different arena*—identical marginal rate, burst structure and inter-pulse statistics, no contingency with the receiver’s actual world—changes neither party’s return. The same holds for temporal scrambling, for identity scrambling, and for phantom pulses injected at rates from 0.05 to 0.6 (Fig. 5C). Because a non-significant difference is not evidence of no difference, we report these as two one-sided tests against an equivalence margin fixed in advance at 5 % of the intact condition’s own return: no comparison clears the margin in either direction —

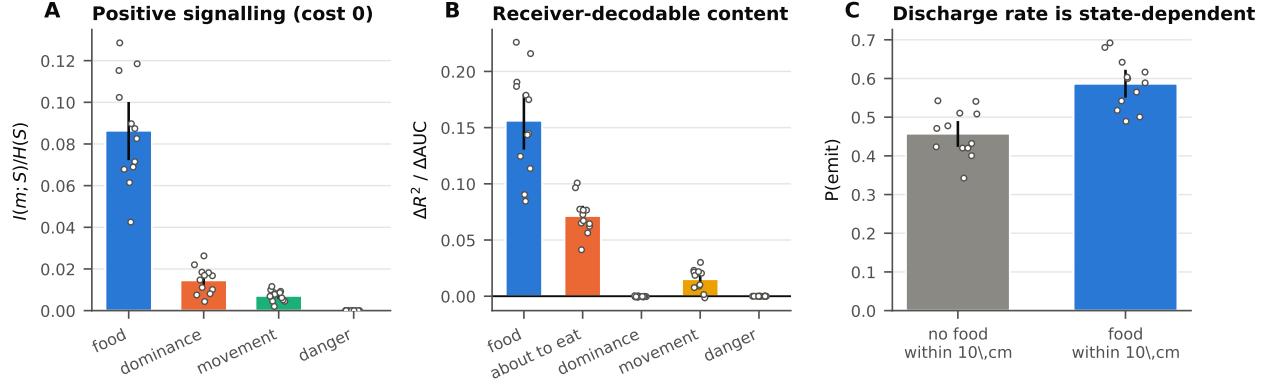


Figure 4: What the pulse train carries. (A) Positive signalling, conditioned on distance-to-neighbour decile. (B) Cross-validated decoding from receiver-observable pulse counts only, minus the time-shifted baseline. (C) Discharge probability by food proximity.

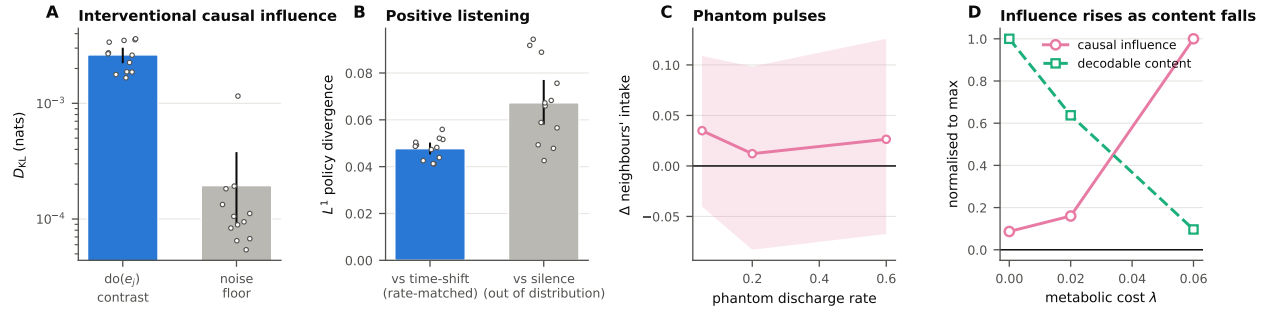


Figure 5: Positive listening, and why it is not enough. (A) Interventional causal influence against its noise floor (log scale). (B) Positive listening against a rate-matched time-shift null and against silence, which is out of distribution. (C) Phantom-pulse dose-response on neighbours' intake. (D) The dissociation: as metabolic cost rises, causal influence per discharge *increases* while the information a receiver can decode *falls*. Influence and informativeness move in opposite directions, so neither substitutes for the other.

the intervals are simply wider than the ± 2.13 margin (cross-arena replay, $+0.77$ (90% CI $[-2.53, +4.09]$); temporal scrambling, $+0.35$ (90% CI $[-2.96, +3.68]$); removing the public channel entirely, -0.75 (90% CI $[-4.11, +2.57]$)). We therefore report these as *no effect detectable at this sample size*, and explicitly not as demonstrated equivalence. The honest statement is that the design rules out payoff effects larger than roughly a tenth of the intact return, and is not powered to rule out smaller ones.

The contingency of the public channel, as opposed to its presence, is worth approximately nothing to anybody. A receiver can read the emitter's situation off its pulse train, and does adjust its policy when the train changes, but nothing downstream of that adjustment shows up in what either fish eats.

2.6 Cost and predation regulate the probe without creating a signal

The reference model gives discharges no cost. Adding one produces a clean dose-response: discharge probability falls monotonically from 0.495 at $\lambda = 0$ to 0.032 at $\lambda = 0.08$ (Fig. 8). Eavesdropping predation pushes it lower again at every cost, and mortality falls with it, so crypsis works. Cooperative harvesting sustains a higher rate than competition at matched cost.

We expected this to make the pulse train informative. A budget has to be spent where it pays, and a schedule that allocates pulses to the moments that matter should be more diagnostic of the emitter's state than a constant one. The opposite happens. Information about food falls monotonically with cost, from $\Delta R^2 = 0.098$ at $\lambda = 0$ to 0.010 at $\lambda = 0.08$, and the same decline appears for every other target we decoded. It survives normalising per pulse: informativeness per discharge falls from 0.127 to 0.087. The learned response to a metabolic price is a lower baseline rate, not a smarter schedule.

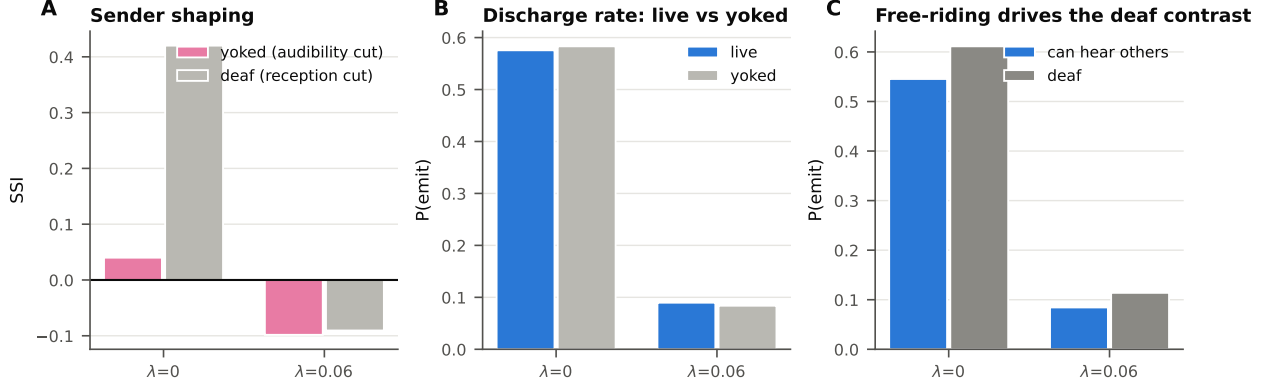


Figure 6: Sender shaping measured with reception held fixed. (A) SSI from the yoked contrast, which cuts audibility only, beside the confounded deaf contrast, which cuts reception. (B) Discharge rate, live versus yoked. (C) The free-riding effect that drives the deaf contrast: fish that can hear neighbours discharge less.

Silence behaves the same way. Discharge probability is essentially identical whether or not a predator is within 25 cm (difference -0.004 , 95% CI $[-0.007, +0.000]$), and danger is close to undecodable from the public channel ($+0.010$ AUC above the time-shifted baseline). Agents manage their average exposure rather than timing their gaps, which is what a prey animal that cannot see the predator coming can do; our predator detects discharges at 60 cm but is itself passively detectable only at about 25 cm.

2.7 Sender shaping, with reception held fixed

Our first attempt at the production-side test compared populations trained with and without knollenorgans, and gave SSI = 0.420 at zero cost. That contrast is confounded, and in a way worth spelling out because it is easy to make. Disabling knollenorgans removes *reception*. It therefore asks whether a fish behaves differently when it can hear others, not whether it behaves differently when others can hear *it*. The direction of the effect gives the confound away: hearing populations discharge *less* than deaf ones (0.545 versus 0.611), which is free-riding—a fish that can hear its neighbours’ probes cuts its own costly sampling—and not audience design, which would push emission up or restructure it.

The yoked condition holds reception fixed and cuts audibility instead. Receivers get knollen input with matched statistics, drawn from another arena, so no sender’s pulses reach anybody while every receiver still has a realistic social input stream. Measured this way, sender shaping is 0.040 at zero cost (Fig. 6A), and discharge rate is 0.576 live versus 0.583 yoked. Emission policy is therefore no more different between the live and yoked worlds than between seeds of the same world. Holding reception fixed, being listened to leaves no detectable trace on how the discharge is scheduled, which is what it means for the pulse to be a cue rather than a signal.

2.8 A decoupled variable is what lets a social function appear

A negative result is only as good as the positive control beside it, and to this point every manipulation has left the discharge welded to its sensory function. We therefore gave the agents a second variable: a discharge subtype, one bit, that rides on the pulse—so sending still costs a pulse and still exposes the fish to predators—but whose value changes nothing about the emitter’s own electric image, nothing about how the pulse illuminates neighbours, and nothing about detectability. Only the content is decoupled, not the act.

Agents mark 52% of their discharges with the subtype, which is indistinguishable from the 50% an unmodulated bit would give, and it carries 0.0158 bits about food proximity above its permutation null. Deleting the subtype leaves the pulse, and therefore every sensory consequence, completely intact, so any payoff change is attributable to content alone. Across the four ecological settings: standard $\times$ compet., $+0.51$ (95% CI $[-0.45, +1.40]$, $p = 0.292$); standard $\times$ cooper., $+0.54$ (95% CI $[-1.81, +2.99]$, $p = 0.640$); sparse $\times$ compet., -1.73 (95% CI $[-2.77, -0.51]$, $p = 0.016$); sparse $\times$ cooper., -0.90 (95% CI $[-2.56, +1.57]$, $p = 0.549$). No cell survives correction for multiple comparisons. Freeing the content of a discharge from its

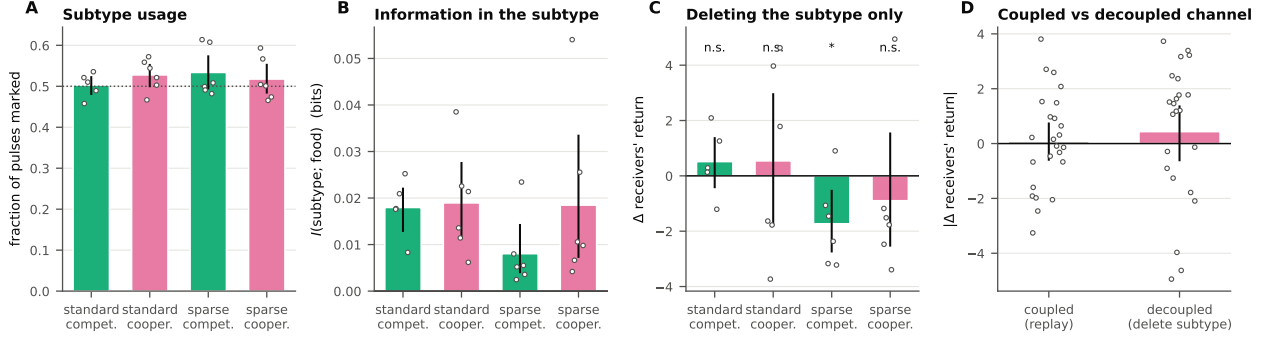


Figure 7: The positive control. A discharge subtype decoupled from sensing, crossed with competitive and cooperative harvests and with a sparse-patch task in which finding food is the bottleneck. (C) deletes the subtype while leaving the pulse, and therefore all sensing, completely intact—the manipulation the coupled channel does not admit.

sensory function was not, on its own, sufficient to make that content payoff-relevant under any pressure we applied.

2.9 The regime map

Fig. 8 shows the full grid, and Fig. 9 the rule-based regime labels. The structure is real and orderly, but it is the structure of a probe schedule being turned up and down, not of a signalling system appearing and disappearing: the axes that move are rate and mortality, while the informational and payoff-relevance axes stay near zero throughout.

3 Discussion

An effect on a receiver is not a signal. The operational definition used by Singh et al. [6]—an EOD that conspecifics detect and that changes their behaviour—is satisfied in our model at every metabolic cost we tested, including zero. It is satisfied because the knollenorgan channel really does carry information and receivers really do respond to it. Yet the same channel has no detectable consequence for anyone’s payoff: replaying, scrambling or fabricating it leaves both parties’ returns inside a 5% equivalence margin. A system can pass every standard diagnostic for communication and still not be communicating.

Establishing the production-side half of the argument turned out to need more care than we first gave it. Disabling knollenorgans removes *reception*, so a hearing/deaf contrast asks whether a fish behaves differently when it can hear others—not whether it behaves differently when others can hear *it*. Those are different questions, and the first is contaminated by free-riding: a fish that can hear its neighbours’ probes cuts its own costly sampling, which is what we in fact observe. The control that isolates being listened to has to hold reception fixed and cut audibility, which is what our yoked condition does.

The muting confound. Silencing a fish is the natural intervention and it is the one available in real experiments, but it removes three things simultaneously. Our decomposition shows that the three shares are not merely different in size—they act on different individuals and in different directions. Losing refference hurts the muted fish; losing the illumination it provided hurts its neighbours; losing its detectability changes how the group is spaced. A single number attributed to “communication” averages over all three. The same caution applies to real ablation work: an EOD-silenced *Gnathonemus* is not a fish with its telephone taken away, it is a fish that has been blinded and has stopped lighting the room for everyone else.

Cost does not manufacture semantics. It is natural to expect that pricing a pulse would turn a probe schedule into an informative one: a budget has to be spent where it pays, and that allocation should make the train diagnostic of the emitter’s state. That is not what happens. Across the whole cost range the pulse train becomes rarer and *less* informative about every target we decoded, and it stays less informative

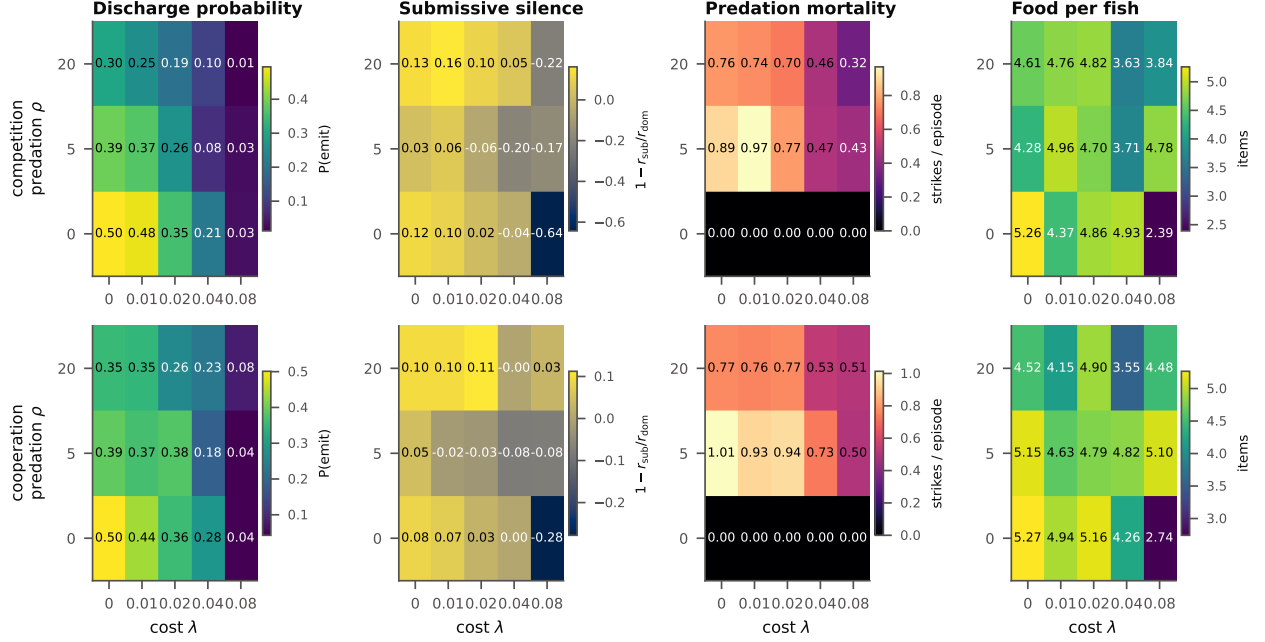


Figure 8: Metabolic cost and eavesdropping predation jointly set discharge policy. Rows are competitive and cooperative harvests; each cell is a mean over independent seeds.

after normalising per pulse. The learned response to a price is a lower baseline rate, not a smarter schedule. Whatever costly signalling does in the classical models [10–12], it presupposes a signal whose honesty is at stake; it does not by itself bring one into being.

What we should not conclude is that a binary pulse leaves no room for a code. A binary event repeated in continuous time supports a large space of temporal codes—inter-pulse intervals, bursts, pauses, event-locked patterns—and mormyrid communication is known to use pulse timing and waveform identity [3, 13, 14]. Our agents *failed to exploit* the temporal code available to them under the pressures we applied. That is a claim about what learning found, not about what the channel permits.

Silence tracks the regime, not the moment. Cessation of discharge is well documented in mormyrids, both as a predator-avoidance response and as a submissive display [15, 16], and we expected the model to reproduce a momentary version of it: quiet when the predator is close, loud when it is far. It does not. Discharge probability under predation is essentially the same whether or not a predator is within 25 cm (-0.004 , 95% CI $[-0.007, +0.000]$), and danger is close to undecodable from the public channel (0.010 above the time-shifted baseline in AUC). What the agents learn is a lower *baseline* rate wherever predators exist at all, and that suffices—mortality falls with it, so crypsis works without any moment-to-moment timing.

This is a substantive difference from the biology, and the likely reason is sensory: our predator becomes passively detectable only at about 25 cm while it detects discharges out to 60 cm, so a fish usually has no warning to respond to. A prey animal that cannot see the predator coming can only manage its average exposure. Testing the momentary version of the behaviour would need a longer-range warning cue.

Signalling as a public good and a public hazard. Two externalities operate in opposite directions and neither is in the reward function. A discharge illuminates neighbours’ worlds, which helps them; it also jams their passive ampullary sense and draws an electroreceptive predator toward the whole group, which harms them. Cooperative harvesting sustains a higher discharge rate than competition at matched cost, which is consistent with the public-good component being internalised when the harvest is shared.

We are careful not to state that as a mechanism. Our channel interventions found no detectable payoff consequence of the public channel, so we cannot simultaneously claim that neighbours benefit enough from a pulse to explain the rate difference. The higher cooperative rate may instead reflect altered movement and

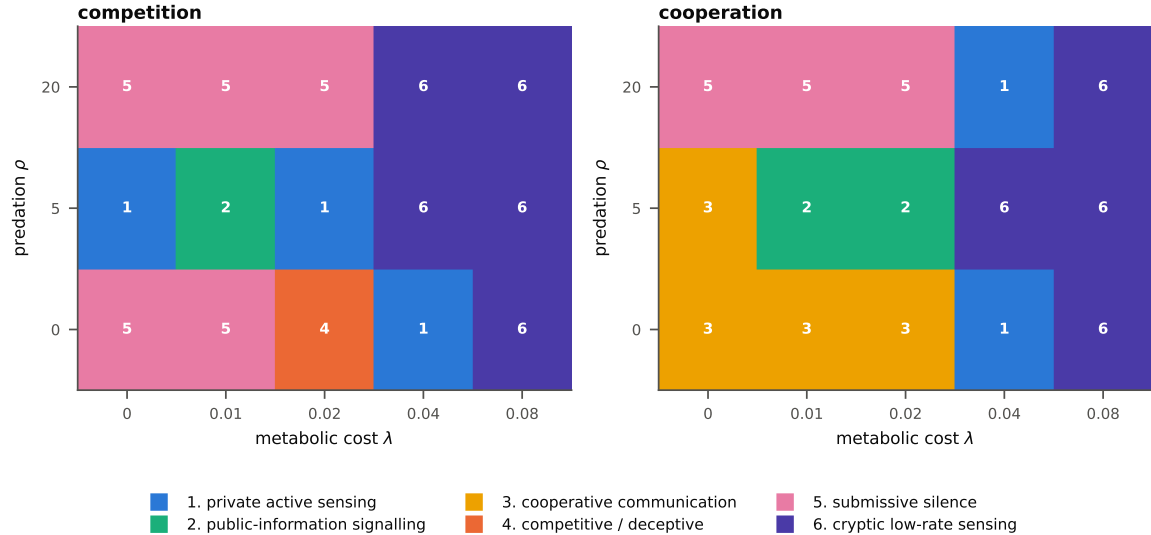


Figure 9: Regimes. Rule-based labels from the pooled metric vector; thresholds are percentiles of each metric’s own distribution across the grid, not hand-set. Boundaries describe a continuum, not sharp transitions.

encounter distributions, different private sensing requirements under a shared harvest, or a social effect below our resolution. Establishing the public-good account would require showing causally that the rate difference depends on neighbours receiving the pulses, and that experiment is not in this paper.

Relation to emergent-communication work. Most emergent-communication studies equip agents with a dedicated, costless, otherwise-useless channel [17–19]. Under those conditions any use of the channel is communicative by construction, and the interesting question is what code emerges. Electric fish invert the setup: the channel is a physical act with a large private benefit, and the interesting question is whether *any* of its use is communicative. We think this is the more common situation in biology—most signals are thought to be ritualised from behaviours that originally had other functions—and it is a setting in which the standard metrics need the production-side test that SSI supplies. Positive signalling and positive listening are both necessary; neither, nor both together, is sufficient.

4 Limitations

We list these plainly because several of them bound the claims above.

1. **Learning is not evolution.** Our populations adapt within a training run by policy-gradient learning under a fixed reward. The Scott-Phillips criterion refers to evolutionary history, and SSI is a within-lifetime analogue: it asks whether the emission policy that learning arrives at depends on whether receivers respond. A genuine evolutionary outer loop over heritable emission traits would test the criterion directly, and we did not run one. We write “emerge under learning” throughout rather than “evolve”.
2. **One arena scale, one group size.** Everything is measured with four fish in a 60×60 cm arena, in which the knollenorgan channel at unit gain reaches every conspecific. The range sweep partially addresses this, but group size is not varied, and the upstream model randomises arena size in a way we deliberately do not.
3. **Pulses are binary and fixed-amplitude.** Mormyrid electric communication uses inter-pulse-interval patterning and species- and individual-specific waveform structure [13, 14]; in wave-type gymnotiforms, amplitude modulations carry honest size information [20]. Our knollenorgan channel is all-or-none, as upstream, so waveform and amplitude signalling are structurally impossible in the coupled condition. This is precisely the limitation the decoupled condition is designed to lift, and it lifts only the crudest version of it: one bit of subtype, not a graded waveform.

4. **Episodes are short.** 512 steps is about 6 s of fish time. Dominance relationships in real fish are established over much longer horizons, so the dominance-related results should be read as within-encounter, not as stable social structure.
5. **The metabolic cost is a reward term, not a physiological model.** We express cost in reward units and report the fraction of the achieved return it consumes so that it can be compared with measured energy budgets, but we do not model the electrocyte physiology that sets the true cost, nor its dependence on amplitude.
6. **The predator is a caricature.** It is guided by discharges alone. Real electroreceptive predators also use vision, mechanoreception and olfaction, so real silence buys less safety than it does here, and the cryptic regime is correspondingly easier to reach in our model than in a river.
7. **Regime labels are a classification imposed on a continuum.** The phase diagram is generated by thresholding continuous metrics at percentiles of their own pooled distribution. The boundaries are therefore descriptive, not sharp transitions, and we make no claim about criticality or bifurcation.

5 Methods

5.1 Simulator

The environment is a GPU reimplementation of the weakly-electric-fish simulator of Singh et al. [6], rewritten so that thousands of independent arenas are advanced as a single set of tensor operations. All electrostatic constants, receptor counts, receptor placements, dynamic ranges, transduction functions, locomotor limits and reward coefficients are transcribed from that codebase; the differences are enumerated in §5.6.

State and kinematics. Each arena holds $F = 4$ fish, $N = 48$ food items and optionally one predator, in a $60 \text{ cm} \times 60 \text{ cm}$ rectangular tank with insulating walls. Fish i has position $\mathbf{x}_i$, heading θ_i and a body size $s_i \in [0, 1]$ that is fixed within an episode and sets a dominance rank. The timestep is $\Delta t = 1/83 \text{ s} = 12.05 \text{ ms}$, matched by the original authors to the social echo-response latency of *Gnathonemus petersii* [21]; episodes last 512 steps ($\approx 6.2 \text{ s}$). Motion is first-order in the commanded velocities,

$$\omega_i = u_i^{\text{turn}} \omega_{\max}(1 + s_i), \quad \mathbf{x}_i \leftarrow \mathbf{x}_i + \hat{u}(\theta_i) \max(u_i^{\text{fwd}}, 0) v_{\max}(1 + s_i), \quad (1)$$

with $v_{\max} = 0.422 \text{ cm per step}$ and $\omega_{\max} = 0.0434 \text{ rad per step}$, taken from the empirical percentiles used upstream. A fish whose proposed position would overlap another fish does not move and receives a collision penalty; walls clamp position without penalty.

Electric fields. All fields are exact Coulomb superposition, with the distance softening $\varepsilon_m = 10^{-5} \text{ m}$ applied *before* exponentiation exactly as upstream. A discharging fish carries two monopoles at $\pm 0.5 \text{ cm}$ along its body axis with charges $\pm q_0$, $q_0 = 1.11 \times 10^{-15} \text{ C}$ [22]:

$$\mathbf{E}_{\text{mono}}(\mathbf{r}) = k \sum_p \frac{q_p(\mathbf{r} - \mathbf{r}_p)}{(\|\mathbf{r} - \mathbf{r}_p\| + \varepsilon_m)^3}, \quad \mathbf{E}_{\text{dip}}(\mathbf{r}) = k \sum_d \left[\frac{3(\mathbf{p}_d \cdot \mathbf{u}_d)\mathbf{u}_d}{\|\mathbf{u}_d\|^5} - \frac{\mathbf{p}_d}{\|\mathbf{u}_d\|^3} \right], \quad (2)$$

with $k = 8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$ and $\mathbf{u}_d = \mathbf{r} - \mathbf{r}_d$. Fish and food carry permanent intrinsic dipoles (1.11×10^{-23} and $1.11 \times 10^{-24} \text{ C m}$). Food items and non-discharging fish are polarisable insulating spheres,

$$\mathbf{p}_n^{\text{ind}} = 3\varepsilon_0 V_n \chi \mathbf{E}(\mathbf{y}_n), \quad V_n = \frac{4}{3}\pi a_n^3, \quad \chi = -0.5, \quad (3)$$

with fish moments capped at $q_0 a_n$. Walls contribute one order of images across each of the four faces, unattenuated and without sign flip.

Receptors. Each fish carries 36 mormyromasts (10 concentrated in a $\pi/3$ chin fovea, 26 spread over the body), 24 ampullary organs and 12 knollenorgans *per conspecific*. A receptor reads the projection of the local field onto its outward normal and transduces it by a sign-preserving logarithmic compression into $[-1, 1]$,

$$\hat{s} = \text{sgn}(s) \frac{\log_{10} \text{clip}(|s|, E_{\min}, E_{\max}) - \log_{10} E_{\min}}{\log_{10} E_{\max} - \log_{10} E_{\min}}. \quad (4)$$

Mormyromasts use $[E_{\min}, E_{\max}] = [5 \times 10^{-8}, 5 \times 10^{-2}] \text{ V m}^{-1}$ and report the field of the *induced* dipoles alone: the direct discharge field is cancelled, so what remains is the electric image of objects. Images carried by a conspecific’s discharge rather than one’s own are rescaled by 100, as upstream. Ampullary organs ($[2 \times 10^{-10}, 2 \times 10^{-8}] \text{ V m}^{-1}$) sense only the permanent intrinsic dipoles and are therefore independent of any discharge; a conspecific discharge degrades them (multiplicative noise fraction rising from 0.05 to 0.5), so emitting jams a neighbour’s passive sense. Knollenorgans are all-or-none: a slot fires with the sign of the local field when $|s| > 2 \times 10^{-7} \text{ V m}^{-1}$ (a detection range of $\approx 100 \text{ cm}$ at unit gain). Crucially, the knollenorgan array is indexed *by emitter*, so sender identity and bearing are preserved, and each slot carries a noisy scalar of the emitter’s relative body size.

Task and reward. Food is distributed in four Gaussian patches ($\sigma = 6 \text{ cm}$) and is not replenished. A fish eats an item within 2 cm and $\pm\pi/4$ of its heading; contested items go to the larger fish. The per-step reward is

$$r_i = 10 n_i^{\text{eat}} + \frac{d_i^{\text{prev}} - d_i^{\text{food}}}{W+H} - 5 b_i^{\text{bitten}} (1 + s_{\text{biter}} - s_i) - 0.001 b_i^{\text{bite}} - 0.5 c_i^{\text{coll}} \underbrace{- \lambda e_i}_{\text{new}} \underbrace{- \rho k_i}_{\text{new}}, \quad (5)$$

where $e_i \in \{0, 1\}$ indicates a discharge and k_i a predator strike. The first five terms are upstream’s; λ (metabolic cost) and ρ (predation) are introduced here. Under cooperative harvesting a fraction α of each neighbour’s intake is added to r_i .

Predators. A predator perceives only fish that discharged on the current step and lie within 60 cm, weights them by $1/(d+1)$, and advances 0.45 cm toward the loudest. Within 4 cm it strikes the nearest fish—not necessarily the emitter—and enters a 40-step refractory period before respawning at a random location. A silent fish is invisible, but a silent fish beside a discharging neighbour is not safe, so signalling imposes a risk externality on the group. The predator carries a strong intrinsic dipole and is therefore detectable passively at $\approx 25 \text{ cm}$, far inside the range at which it detects discharges.

5.2 The channel algebra

A discharge by fish j at time t is routed through three independent binary masks:

V_j^{self}	does j ’s pulse produce j ’s own electric image?
V_j^{illum}	does j ’s pulse produce images on <i>other</i> fish’s mormyromasts?
V_j^{det}	does j ’s pulse register on other fish’s knollenorgans?

Setting all three equal to the emitted action recovers the biological default. The conditions used here are: PRIVATE ($V^{\text{illum}} = V^{\text{det}} = 0$), CUE-ONLY ($V^{\text{det}} = 0$), SIGNAL-ONLY ($V^{\text{illum}} = 0$), SOCIAL-ONLY ($V^{\text{self}} = 0$), and MUTE (all zero, the upstream intervention). Three further manipulations act on what receivers hear without changing what the emitter does, so that any change in receiver behaviour is attributable to the public channel alone:

- **Phantom.** Receivers perceive discharges from j drawn from an exogenous Bernoulli process that j did not choose.
- **Replay.** j ’s public train is replaced by the train the same policy produced in a *different arena*. Marginal rate, burst structure and inter-pulse-interval statistics are preserved exactly—these are the same trains—while the contingency between a pulse and the receiver’s current world is destroyed. A context-swapped variant pairs each arena with the one whose food-proximity time series is most opposite.
- **Scrambling.** Timing scrambling circularly shifts each train, preserving rate and destroying temporal alignment; identity scrambling permutes the knollenorgan slots, preserving rate and timing and destroying sender identity.

5.3 Learning

Agents are trained with recurrent multi-agent PPO [23, 24] under centralised training with decentralised execution: one shared actor (2-layer MLP encoder, LayerNorm, tanh, 128 units $\rightarrow$ GRU with 128 units [25]) and a centralised critic that observes all agents’ concatenated observations. The action space is hybrid—two Gaussians for thrust and turn, two Bernoullis for discharge and bite. Keeping the discharge decision explicitly Bernoulli matters for the analysis, because its probability is exactly the quantity we intervene on. GAE [26] with $\gamma = 0.99$, $\lambda = 0.95$; PPO clip 0.2; 4 epochs, 4 minibatches; Adam at 3×10^{-4} ; 512 parallel arenas; 64-step truncated BPTT; 18×10^6 agent-steps per run. Every condition is trained from scratch with independent seeds; *no seed selection of any kind is applied*, and all reported statistics pool over all seeds.

5.4 Measurement

Interventional causal influence. Jaques et al. [8] define the influence of agent j on agent i as the KL divergence between i ’s policy conditioned on j ’s actual action and its policy marginalised over j ’s alternatives, and estimate it with a learned model of other agents. Owing the simulator, we evaluate the counterfactual directly:

$$\text{CIE}_t^{j \rightarrow i} = D_{\text{KL}}[\pi_i(\cdot \mid h_t^i, o_t^i[\text{do}(e_j=1)]) \parallel \pi_i(\cdot \mid h_t^i, o_t^i[\text{do}(e_j=0)])], \quad (6)$$

where $o_t^i[\cdot]$ denotes re-rendering i ’s entire sensory input under the intervention with the recurrent state h_t^i and every other cause held on its factual path. The KL is analytic for the factored Gaussian $\times$ Bernoulli policy. The null is the same quantity computed between two re-renderings of the *same* intervention under independent sensory-noise draws: the divergence the policy shows when nothing about the message changed.

Positive signalling and positive listening. Positive signalling [7] is the mutual information between a window descriptor of the emitter’s pulse train (discharge count over 21 steps, quantised) and the emitter’s private state, normalised by the state entropy and computed *conditionally on* distance-to-nearest-neighbour decile. The stratification is essential: in a patchy world a fish near a neighbour is also near food, so an unconditioned mutual information can be manufactured entirely by geometry. Estimation is plug-in with Miller–Madow bias correction, with a permutation null. Positive listening is the L^1 divergence between the receiver’s factual policy and its policy under a time-shifted message, which preserves the marginal rate and destroys contingency—the null recommended by Lowe et al. [7] in preference to a silent channel, which is out of distribution.

Message content. What a receiver could recover is decoded *only* from receiver-observable quantities: the number of pulses it heard from that emitter in a trailing window and whether it heard anything. Ground-truth state is never given to the decoder. Targets are the number of live food items within 10 cm of the emitter, whether the emitter eats within the next 41 steps, emitter body size (dominance), emitter turn command (intended movement), and whether a predator is within 25 cm (danger). We report cross-validated R^2 or AUC minus the same quantity computed on time-shifted messages, so that any score attributable to marginal rate alone is subtracted off.

Sender Shaping Index. A signal, on Scott-Phillips’s definition, requires that production be shaped by its effect on receivers. We train matched populations in a *hearing* world and a *deaf* world (knollenorgans disabled throughout training) and compare their emission policies on a common, condition-neutral batch of observations:

$$\text{SSI} = \frac{\mathbb{E}_c D_{\text{JS}}[\pi_{\text{emit}}^{\mathcal{H}}(\cdot \mid c) \parallel \pi_{\text{emit}}^{\mathcal{D}}(\cdot \mid c)]}{\mathbb{E}_c D_{\text{JS}}[\pi_{\text{emit}}^{\text{seed } a} \parallel \pi_{\text{emit}}^{\text{seed } b}]} - 1. \quad (7)$$

The denominator is the within-world seed-to-seed divergence, so $\text{SSI} \leq 0$ means the emission policy differs no more between worlds than between seeds of the same world—i.e. nobody listening changes nothing, and the pulse is a *cue*. $\text{SSI} > 0$ means production is shaped by reception.

Payoff consequences and the honesty classification. Every channel condition is evaluated against the intact channel *paired within arena*: the two rollouts share the initial food layout and the sensory noise stream, so differencing removes essentially all layout variance. Confidence intervals are 2000-sample bootstraps over arenas. Comparing the intact channel with a rate-matched replay isolates the payoff value of the signal’s *contingency*. Following Maynard Smith and Harper [4] and Dawkins and Krebs [27], a condition is labelled COMMUNICATION when contingency benefits both parties, MANIPULATION when it benefits the sender at the receiver’s expense, and EXPLOITED CUE when it benefits the receiver at the sender’s expense.

5.5 Validation against the reference implementation

Every stage of the sensory pipeline is checked against the original NumPy code in double precision on random scenes (`scripts/validate_physics.py`); results are in Table 1. Agreement is at floating-point round-off, and the knollenorgan and ampullary pathways agree bit-for-bit.

5.6 Deliberate divergences from the reference implementation

We record these explicitly because they bear on how far our results transfer.

1. **Wall-image amplitude.** Upstream’s configuration file sets a reflection coefficient of 0.95, but `ElectricScene._build_refl` never forwards it, so the code that produced the published results uses unattenuated images. We match the code, not the configuration file. Using 0.95 instead shifts the corollary-discharge baseline by $\sim 1.4 \times 10^{-6}$ relative and changes ampullary readings qualitatively, because those are a small difference of large numbers.
2. **Fixed arena and episode length.** Upstream samples arena dimensions per episode from $[40, 300]$ cm and does not report episode length. We fix a 60×60 cm arena and 512 steps so that conditions in a large factorial sweep are exactly comparable. Our conclusions are therefore about one arena scale.
3. **Collision resolution** is synchronous (all fish propose, then all resolve) rather than in agent order.
4. **Network size.** 128-unit GRU rather than 512, and 18×10^6 agent-steps per run rather than 5×10^6 environment steps, chosen to fit 270 independent training runs into the compute budget.
5. **Sensory noise** is present (multiplicative, fractions given above), following the upstream code; the upstream paper text describes a noiseless model.
6. **No seed selection.** Upstream selects a single seed (S1) by an ethological criterion and reports analyses on it. We pool over all seeds.

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